



Wavepacket Approach in Neutrino Oscillation



Baktiar Wasir Farooq

e-mail: 2330238@kmc.du.ac.in

3rd Year B.Sc. (Hons.) Physics

Kirori Mal College, University of Delhi

I.A.C (Internship - IV) Program,

In Collaboration with Jawaharlal Nehru University, New Delhi

Supervised by:

Internal Supervisor: Prof. Bipin Singh Koranga (Kirori Mal College)

External Supervisor: Prof. Ng. K Francis (Jawaharlal Nehru University)

Outline of Presentation:

I will discuss about the given topics in my presentation:

- Introduction about Neutrinos:
 - History and Origin
 - Neutrino Oscillation
- Neutrino Oscillation Formalism:
 - Two - Flavor Neutrino Oscillation
 - Three - Flavor Neutrino Oscillation
 - Hamiltonian solving for Matter Case
 - CP Violation and Jarlskog
- Wavepacket Approach in Neutrino Oscillation:
 - Reason for the Wavepacket Approach
 - Theory for Vacuum and Matter case
- Theoretical and Experimental Results:
 - MINOS
- *Future Ideas and collaborations*
- *Recent Publications , previous works and arXiv preprints*

Introduction about Neutrinos

1) Beta Decay ($\beta - Decay$): (positive and negative):

Example: ${}^{12}_5B \rightarrow {}^{12}_6C + e^{-}$

$${}_0^1n \rightarrow {}_1^1p + e^{-} \quad (\beta - ve \text{ decay})$$

Example: ${}^{15}_8O \rightarrow {}^{15}_6N + e^{+}$

$${}_1^1p \rightarrow {}_0^1n + e^{+} \quad (\beta + ve \text{ decay})$$

Now, If the energy conservation of a $\beta - ve \text{ decay}$ reaction is considered,

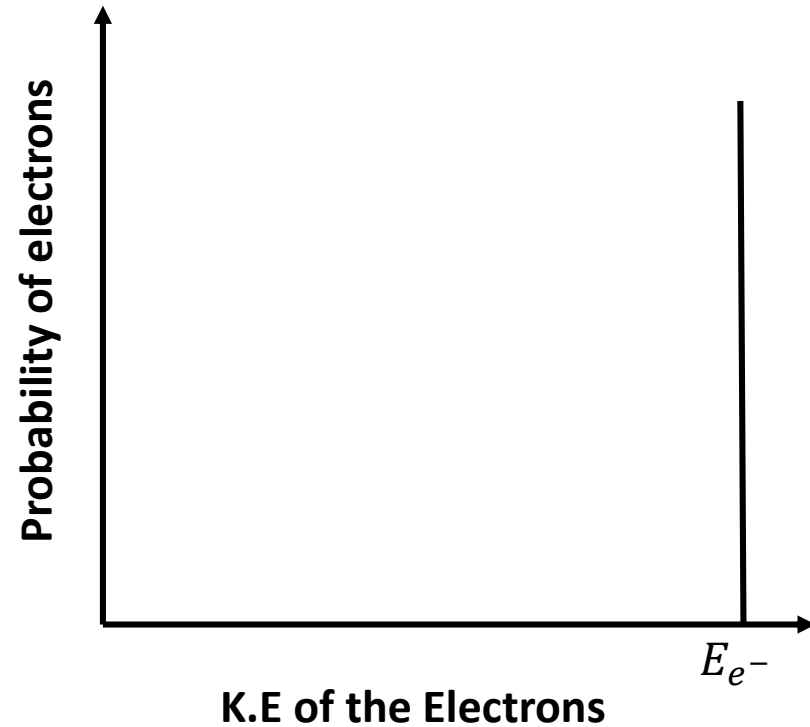
$$E_{e^{-}} = \frac{(m_A^2 + m_e^2 - m_B^2)}{2m_A}$$

Energy is Constant.

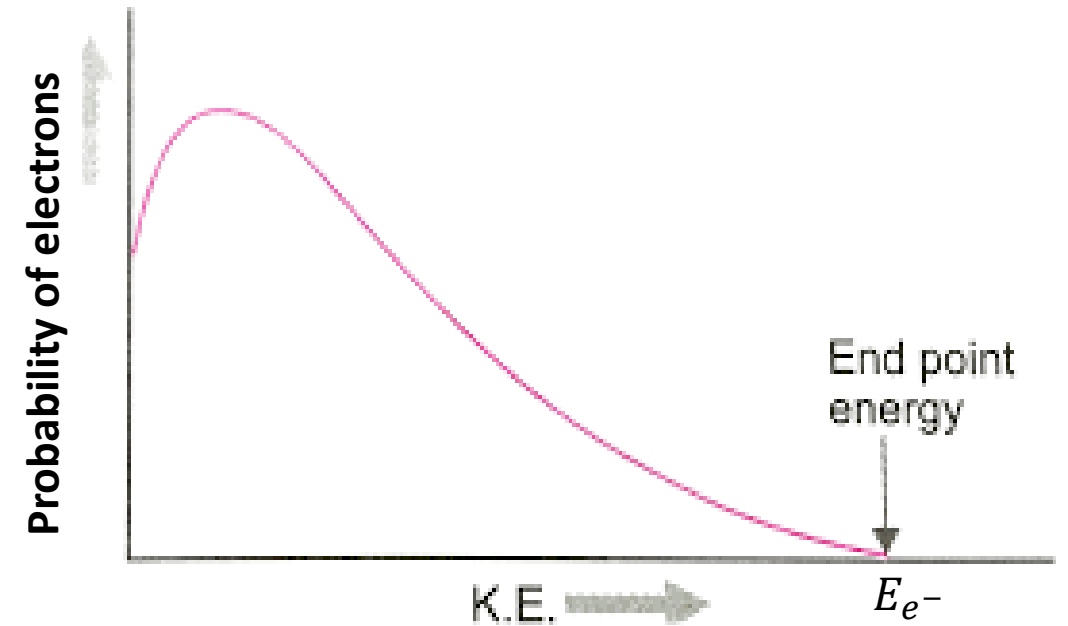


The probability and Energy distribution graph:

Graph associated with the Theoretical result:

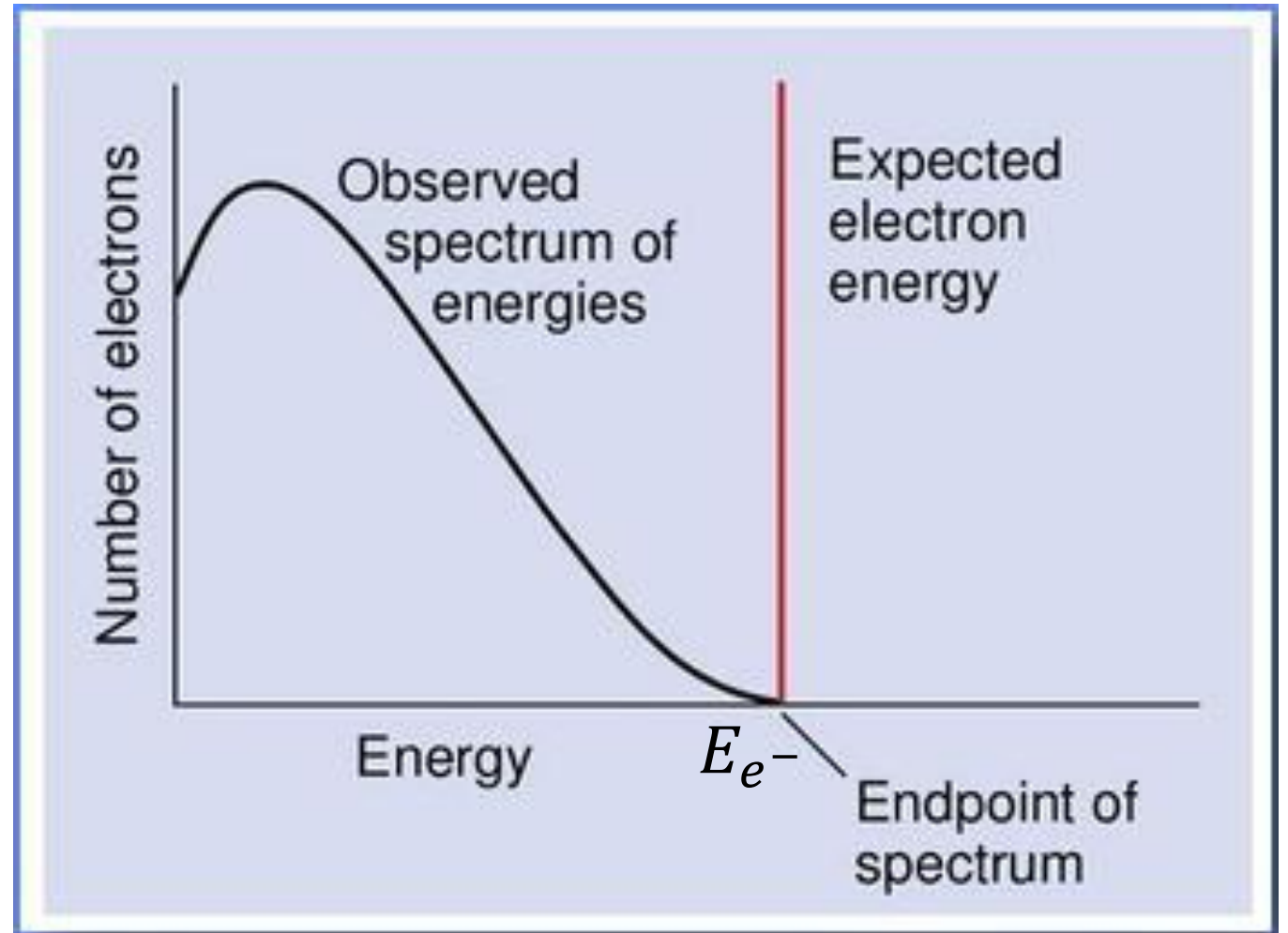


Graph associated with the experimental result:



Comparison Between the Graphs: (Experimental and Theoretical)

- Some energy is missing in the experimental graph for a particular no. of electron which is known as the “missing energy problem”.
- This was mystery for a long time.
- Wolfgang Pauli’s wild card entry in the year 1930, gave the idea of **neutrinos**.



Pauli's Statement. There might be some particle that also release during the β decay reaction which takes the rest of the energy and is known as neutrino and anti-neutrino depending on the type of decay.



Wolfgang Pauli

The symbol of neutrino is a Greek letter (nu): $\bar{\nu}$
 So, the equation of β decay now turns out to be $\rightarrow$

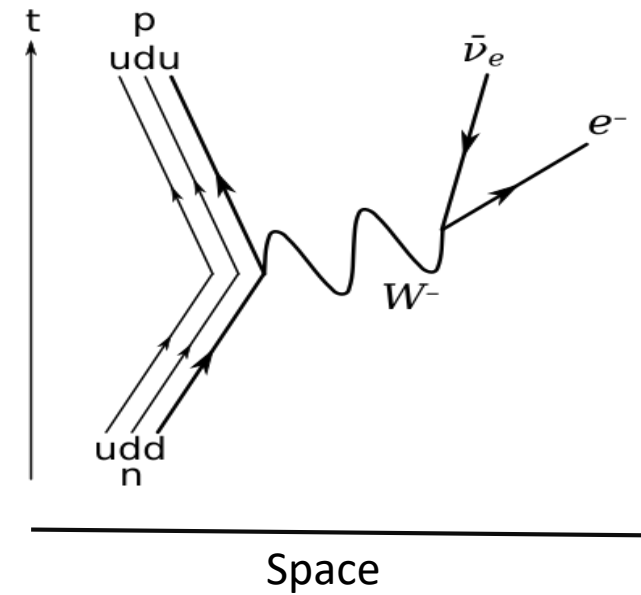
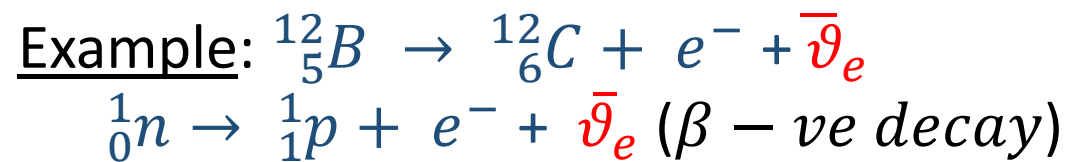


Fig: Feynman diagram of $\beta + ve$ decay.

- b) Example: ${}^{15}_8\text{O} \rightarrow {}^{15}_6\text{N} + e^+ + \nu_e$
- ${}^1_1p \rightarrow {}^1_0n + e^+ + \nu_e$ ($\beta + \nu_e$ decay)

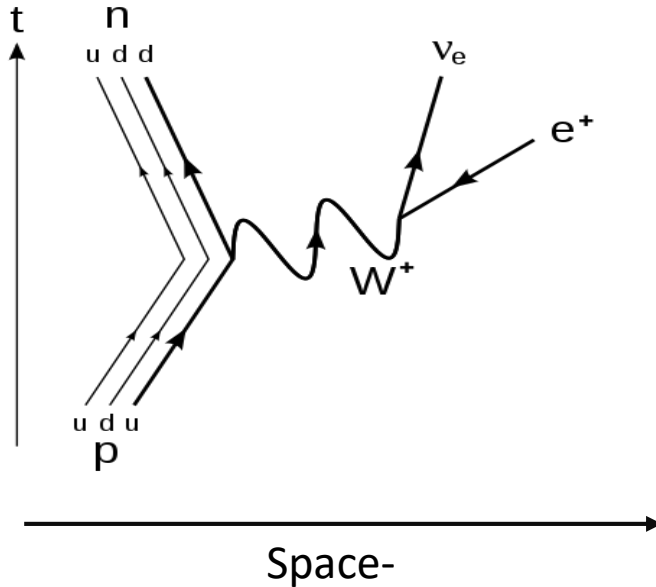
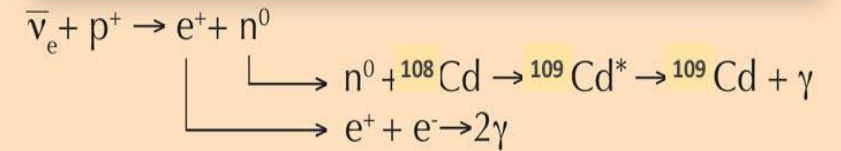
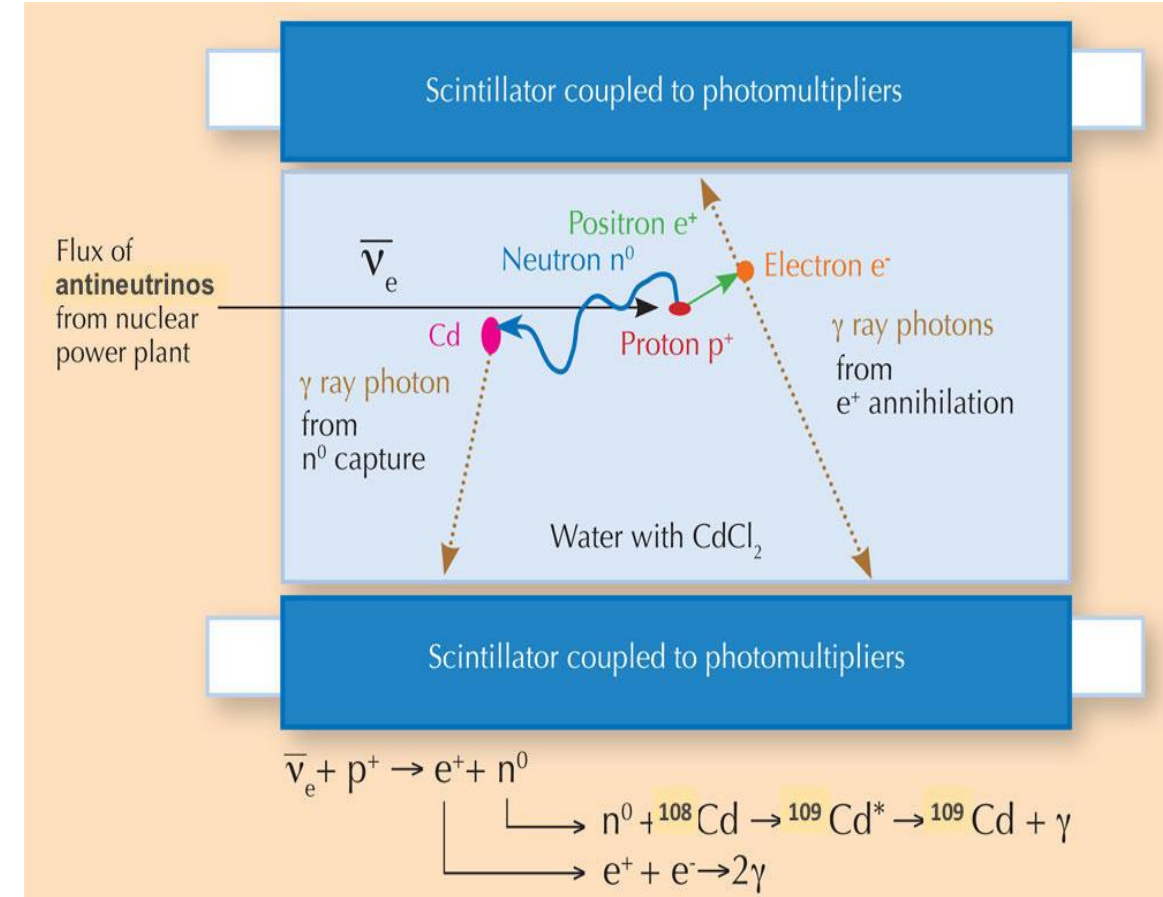


Fig: Feynman diagram of $\beta + \nu_e$ decay.

- This is the actual ***β decay*** reaction and hence, the origin of the neutrinos.

Experimental Evidence of the Neutrinos:

The first experimental evidence of neutrinos came from the "***Cowan-Reines neutrino experiment***" conducted in 1956 by physicists Clyde Cowan and Frederick Reines, where they used a nuclear reactor to produce a large flux of antineutrinos which interacted with protons in a water tank, creating detectable positrons and neutrons, thus confirming the existence of neutrinos; this experiment is considered the definitive proof of the neutrino's existence.



Properties and some facts about Neutrinos:

- 1st property of neutrino is that it is extremely tiny.
- They don't interact with Strong Nuclear Force (S.N.F).
- **Generally, interacts via Weak Nuclear Force (W.N.F).**
- They are neutral.
- There are **3 flavors** of neutrino- *Electron, Muon and Tau neutrino*
(*Existence of Sterile Neutrinos.*)
- They are fermions as it has spin of +1/2.
- Neutrinos are left handed, no right handed neutrinos are found till now and hence, they
violate Parity Conservation.

Helicity: The property of the particle that determines the right and left handedness .

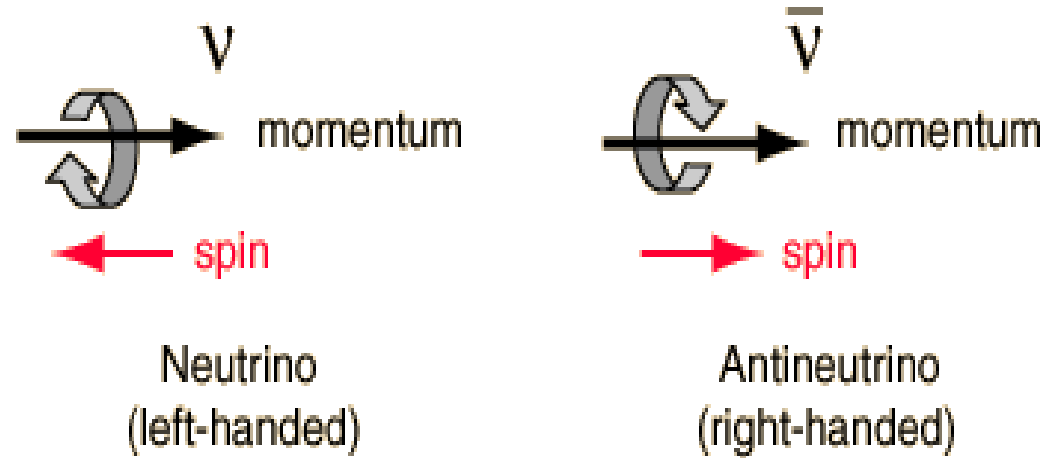


Fig: Systematic representation of the neutrinos and antineutrinos: (left and right handed, respectively)

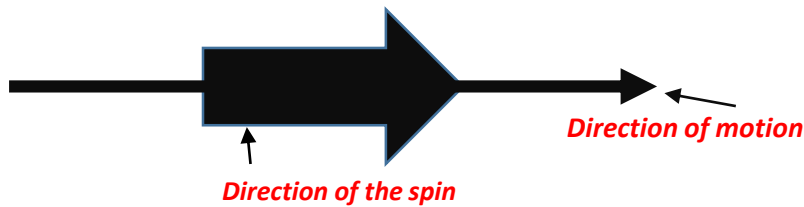


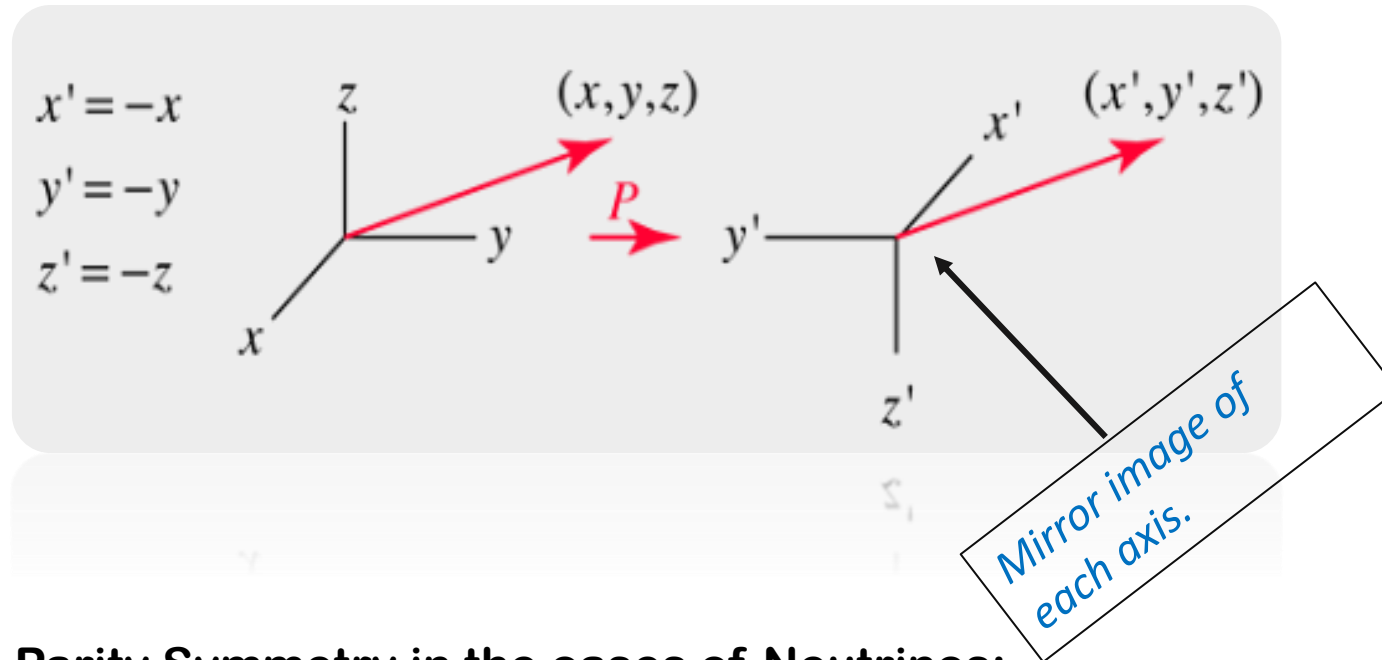
Fig: Final representation of the right handed anti-neutrino

To determine the right and left spin's direction we curl our right hand in the direction of the spin

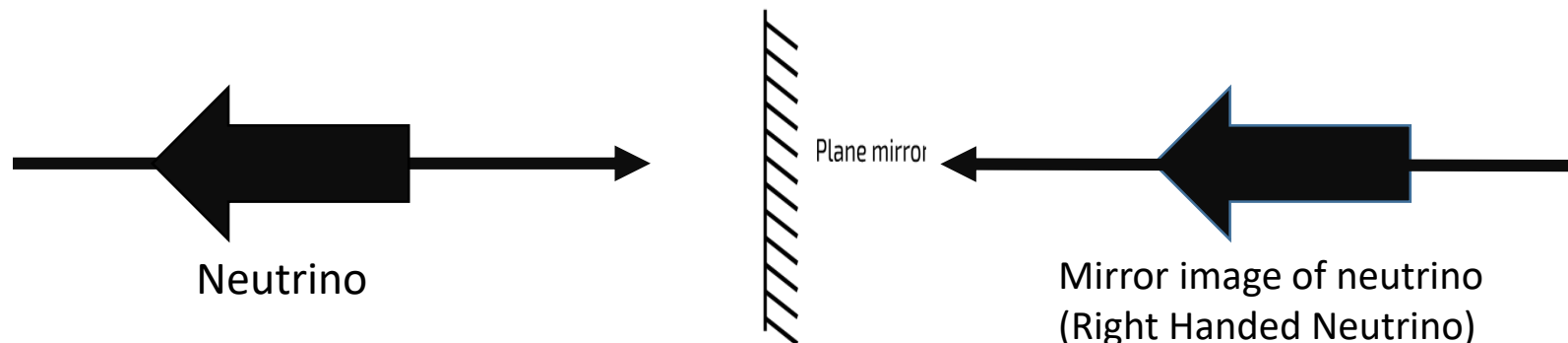


Fig: Final representation of the left handed neutrino

Parity Symmetry: It is a mathematical operation which when carried on a co-ordinate system gets inverted.



Parity Symmetry in the cases of Neutrinos:



For Parity symmetry to be conserved, the mirror image of the neutrino also must be present in the universe but, yet no right handed neutrinos are observed in the nature. A **very interesting experiment** was also being carried out to check the existence of **Right handed neutrinos**.

Neutrino Oscillations

Raymond Davis, a researcher in the **Brookhaven National Laboratory** carried out an experiment to determine the neutrinos during the **solar reactions**. He basically carried out **Homestake experiment** to get the electron neutrino flux. **But there were shocking results!** Only **1/3rd** of the theoretical data was observed from the experimental observations.

Reaction	Frequency	Energy (MeV)	Name
<i>PP I</i>			
$p + p \rightarrow {}^2\text{H} + e^+ + \nu_e$	99.75%	0.0 – 0.42	pp
$p + e^- + p \rightarrow {}^2\text{H} + \nu_e$	0.25%	1.44	pep
${}^2\text{H} + p \rightarrow {}^3\text{He} + \gamma$	100%	—	
${}^3\text{He} + {}^3\text{He} \rightarrow {}^4\text{He} + 2p$	85%	—	
<i>PP II</i>			
${}^3\text{He} + {}^4\text{He} \rightarrow {}^7\text{Be} + \gamma$	15%	—	
$e^- + {}^7\text{Be} \rightarrow {}^7\text{Li} + \nu_e$	99.99%	0.86, 0.38	${}^7\text{Be}$
$p + {}^7\text{Li} \rightarrow {}^4\text{He} + {}^4\text{He}$	100%	—	
<i>PP III</i>			
$p + {}^7\text{Be} \rightarrow {}^8\text{B} + \gamma$	0.01%	—	
${}^8\text{B} \rightarrow {}^4\text{He} + {}^4\text{He} + e^+ + \nu_e$	100%	0–14.1	${}^8\text{B}$

Fig: The pp cycle and the creation of electron neutrinos inside the sun

This was termed as the “solar neutrino problem”

SNU= Solar Neutrino Unit

It's like by using **1 solar neutrino unit = 1 neutrino is being captured in 10^{36} second.**

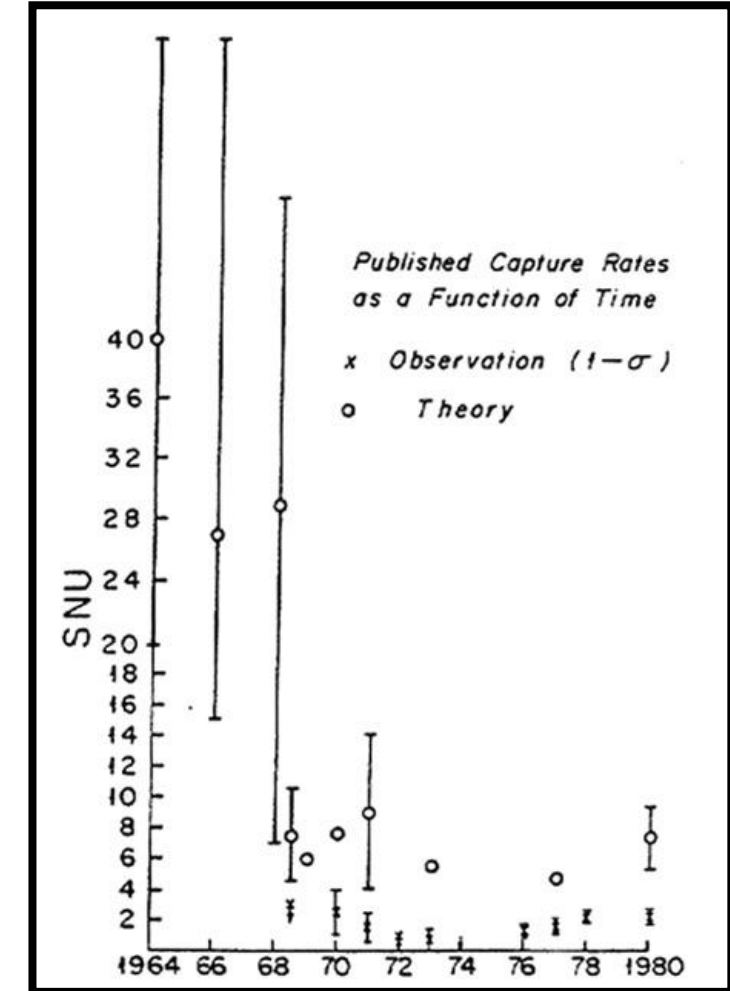


Fig: the plot of the observation of Raymond Davis for a year of (1964-1980)

Bruno Pontecarvo, an Italian scientist gave the idea of **neutrino mixing** – he said that neutrinos can be assumed to be a mixture of certain quantum states and as they travel their identities transform. This was the idea of **Neutrino Oscillation**.

Let's get back the **Solar Neutrino Problem**-

Later, after the discovery of the ϑ_μ , it was found that some of the ϑ_e got converted to ϑ_μ .

This was the experimental evidence of the Neutrino Oscillation.

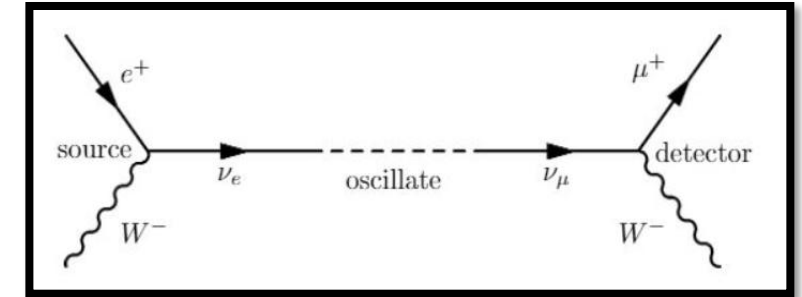


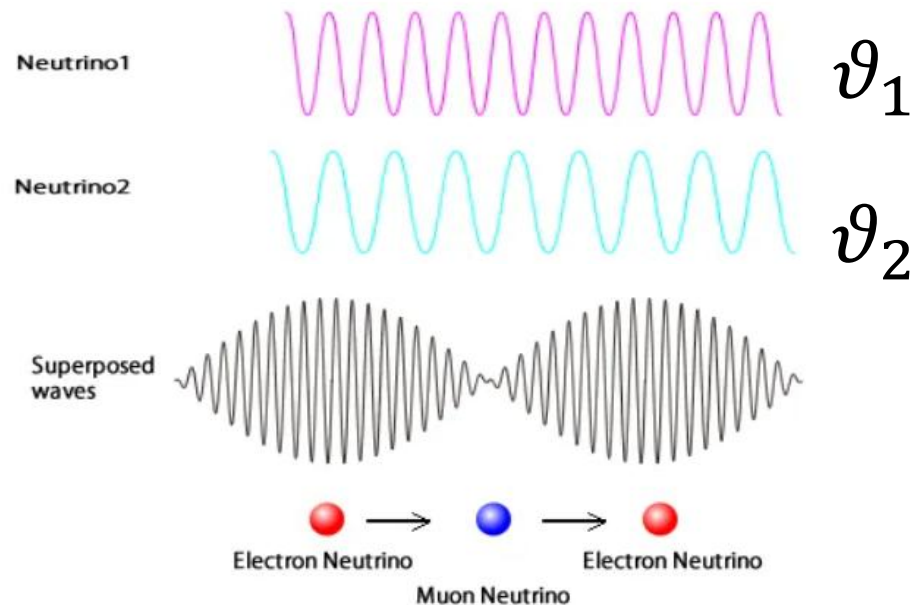
Fig: Feynman diagram for Solar Neutrino Oscillations.

Quantum or Mass Eigen States: These are the waves out of which the neutrinos are made of. (ϑ_1 or ϑ_2)

$$\begin{pmatrix} \vartheta_e \\ \vartheta_\mu \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} \\ U_{\mu1} & U_{\mu2} \end{pmatrix} \begin{pmatrix} \vartheta_1 \\ \vartheta_2 \end{pmatrix}$$

Flavor Eigen States Mixing matrix Mass Eigen States

This is the matrix representation of the 2 flavor neutrinos.



This is the basic idea of what is happening in neutrino oscillation and how is it changing its identity.

Neutrino Oscillation Formalism

- Two-Flavor Neutrino Oscillation:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix}$$

Here, the ν_1 and ν_2 are mass eigen states with masses m_1 and m_2 , respectively, such that $\Delta m \neq 0$ and $\theta \neq 0$.

The expression of Survival Probability: $P_{(\nu_e \rightarrow \nu_e, t)} = 1 - \sin^2 2\theta \sin^2 \left(\frac{(E_2 - E_1)t}{2} \right)$

This is not the final expression we need some modification!

These are the Modifications:

$$E_i = \sqrt{P_i^2 + m_i^2}$$

By using the Binomial Expansion:

$$E_i = P_i + \frac{m_i^2}{2P_i}. \text{ Here, } i=1,2.$$

Using the Equal Momentum Approximation:

$$E_2 - E_1 = \frac{m_2^2 - m_1^2}{2P} \approx \frac{m_2^2 - m_1^2}{2E}$$

Getting back to the phase term: $\frac{E_2 - E_1}{2}t = \frac{\Delta m^2 (eV^2)}{4E(eV)} L(m)$ So, we can see dimensionally that the phase term has unit eV-m! as we know that the Phase is dimensionless and since, $\hbar c = 1.97 \times 10^{-7} eVm$. So, we divide the phase term by $\hbar c$. Which results to: $1.27 \frac{\Delta m^2 (eV^2)}{E(MeV)} L(m)$

Finally we get the expression for the Survival Probability of Neutrino! Which is:

$$P_{(\nu_e \rightarrow \nu_e, t)} = 1 - \sin^2 2\theta \sin^2 \left(1.27 \frac{\Delta m^2 (eV^2) L(m)}{E(MeV)} \right)$$

Here, L is the baseline length in meters.

E is the average energy of the neutrino flux detected by the detector in MeV.

θ is the mixing angle.

- Three-Flavor Neutrino Oscillation:

P.M.N.S Matrix Basic Idea

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = U_{\text{PMNS}} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$

The standard parameterization of the PMNS matrix is:

$$U_{\text{PMNS}} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix}}_{\text{Atmospheric Neutrino Problem}} \underbrace{\begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix}}_{\text{Reactor Neutrino Problem}} \underbrace{\begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{Solar Neutrino Problem}}$$

$$U_{\text{PMNS}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}$$

where $s_{ij} = \sin \theta_{ij}$ and $c_{ij} = \cos \theta_{ij}$.

- **The doubt I had here is – The phase term exists in the 3 flavor case but not in the 2 flavor case. Why so?**
- The solution to that is, in Quantum Mechanics, the mixing matrix is always complex so the matrix which I am using is the modified matrix where we remove the complex term to get the most simplified form.
- For the 2 flavor case, on assuming the mass eigenstates as $\nu_1 \rightarrow e^{i\alpha} \nu_1$, $\nu_2 \rightarrow e^{i\beta} \nu_2$. when we rephrase the mixing matrix, the phase terms can be simplified and we end up with no phase term!! So, no C.P Violation!
- For the 3 flavor case, when we do the similar operation, there is a phase term left which is termed as the C.P Phase (represented by : $\delta_{C.P}$)

This $\delta_{C.P}$ phase is one of the most important factors and one of the greatest puzzles if we try to understand the physics of Neutrino Oscillations.

$$|\nu_\alpha(t)\rangle = \sum_k U_{\alpha k}^* e^{-iE_k t} |\nu_k\rangle.$$

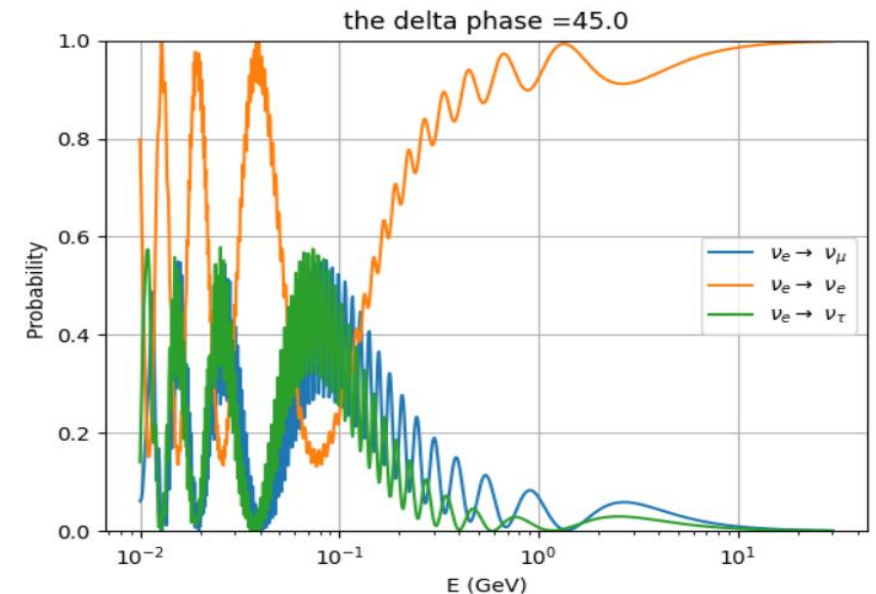
$$P_{\alpha\beta} = \delta_{\alpha\beta} - 4 \sum_{k>j} \text{Re}(U_{\alpha k}^* U_{\beta k} U_{\alpha j} U_{\beta j}^*) \sin^2\left(\frac{\Delta m_{kj}^2 L}{4E}\right) + 2 \sum_{k>j} \text{Im}(U_{\alpha k}^* U_{\beta k} U_{\alpha j} U_{\beta j}^*) \sin\left(\frac{\Delta m_{kj}^2 L}{2E}\right)$$

This is the final formula for Three - Flavor Neutrino Oscillation in the Vacuum case.

The first term represents the flavor conserving component, the second term accounts the flavor oscillations due to the real parts of the mixing elements, and the third term—proportional to the imaginary components of U —encodes the **CP violation** in neutrino oscillations.

I am talking about this term C.P Phase and C.P Violation what does it mean??
I will drive into it right now from the next slide!

My computational analysis in in the variation of C.P Phase for the Vacuum Case in the Three- Flavor Neutrino Oscillation.



- Hamiltonian solving for Matter Case

$$\hat{H}_f^m = U \left[\frac{1}{2E} \begin{pmatrix} 0 & 0 & 0 \\ 0 & \Delta_{21} & 0 \\ 0 & 0 & \Delta_{31} \end{pmatrix} \right] U^\dagger + \begin{pmatrix} V_{cc} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\hat{H}_f^m = W \Lambda W^\dagger \quad \text{where} \quad \Lambda = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

When we try to re write the upper equation in the form of the vacuum case.

$$|\nu_\alpha\rangle = \sum_k W_{\alpha k}^* |\nu_k^m\rangle \quad \text{and} \quad S(L) = e^{-iW\Lambda W^\dagger L}$$

$$P_{\beta \rightarrow \alpha} = \left| \sum_k^3 W_{\beta k} S(L) W_{\alpha k}^* \right|^2$$

This method directly can be used in the computational formalism.

Here, V_{cc} is the matter induced potential (due **to charged current interaction**) $V_{cc} = \sqrt{2}G_F n_e$

where G_F is the Fermi's Constant and n_e is the number density of electron in the earth matter

CC Charged Current Reaction	$\nu_e + d \rightarrow p + p + e^-$
NC Neutral Current Reaction	$\nu_x + d \rightarrow \nu_x + p + n$

$$|H_f^m - \lambda I| = 0 \quad 21$$

$$\lambda^3 - \text{Tr}(H_f^m)\lambda^2 + \frac{1}{2} [\text{Tr}(H)^2 - \text{Tr}(H)^2] \lambda - \det(H_f) = 0 \quad 22$$

$$\lambda_j = \left(-\frac{b}{2} + \sqrt{\left(\frac{b}{2}\right)^2 + \left(\frac{a}{3}\right)^3} \right)^{1/3} \omega^j + \left(-\frac{b}{2} - \sqrt{\left(\frac{b}{2}\right)^2 + \left(\frac{a}{3}\right)^3} \right)^{1/3} \omega^{-j} - \frac{\text{Tr}(H_f^{\text{matter}})}{3}, \quad (23)$$

$$j = 0, 1, 2.$$

In this equation (23), $\omega = e^{i(\frac{2\pi}{3})}$, $a = q - \frac{p^2}{3}$, and $b = \frac{2p^3}{27} - \frac{2qp}{3} + r$. The values of p,q and r can be determined by comparing equation (21) and (22).

For analytical way of getting the eigenvalues, one can use the **cardano change in variables**.

```
for l in range (0,L):
    def fn_rho(l):
        if l < 260:
            return 2.7
        if 260 < l < 520:
            return 2.75
        if 520 < l < 780:
            return 2.85
        if 780 < l < 1040:
            return 2.9
        else:
            return 2.7
```

Results :

arXiv
> hep-ph > arXiv:2602.02578

High Energy Physics - Phenomenology

[Submitted on 31 Jan 2026]

NeutrinoOsc3Flavor: CP Phase Dependence in Three-Flavor Neutrino Oscillations: A Numerical Study in Vacuum and Matter

Baktiar Wasir Farooq, Bipin Singh Koranga, Ansh Prasad, Imran Khan

```
lamdas (analytical)= [np.float64(8.717284238836079e-16), np.float64(4.015412032613798e-14), np.float64(1.0395081791664509e-13)]
lamdas (computational)= [8.71728424e-16 4.01541203e-14 1.03950818e-13]
lamdas (analytical)= [np.float64(8.718754355506513e-16), np.float64(4.014318263182448e-14), np.float64(1.039616085992915e-13)]
lamdas (computational)= [8.71875436e-16 4.01431826e-14 1.03961609e-13]
lamdas (analytical)= [np.float64(8.720224983325364e-16), np.float64(4.0132248553161504e-14), np.float64(1.039723956151726e-13)]
lamdas (computational)= [8.72022498e-16 4.01322486e-14 1.03972396e-13]
```

- Charge Conjugation and Parity Violation (CP Violation)

- C (Charge Conjugation) is the operation that transforms particles into antiparticles; P (is parity), which transforms left-spinning particles to right-spinning ones

$$P_{\nu_\alpha L \rightarrow \nu_\beta L} \xrightarrow{P} P_{\nu_\alpha R \rightarrow \nu_\beta R}$$

$$P_{\nu_\alpha L \rightarrow \nu_\beta L} \xrightarrow{C} P_{\bar{\nu}_\alpha L \rightarrow \bar{\nu}_\beta L}$$

- Hence, we can say that $P_{\nu_\alpha L \rightarrow \nu_\beta L} \xrightarrow{CP} P_{\bar{\nu}_\alpha R \rightarrow \bar{\nu}_\beta R}$.

- Now, the most **IMPORTANT QUESTION** is: **Are these two Oscillating Probabilities the same?**

$$P_{\alpha\beta} = \delta_{\alpha\beta} - 4 \sum_{k>j} \text{Re}(U_{\alpha k}^* U_{\beta k} U_{\alpha j} U_{\beta j}^*) \sin^2\left(\frac{\Delta m_{kj}^2 L}{4E}\right) + 2 \sum_{k>j} \text{Im}(U_{\alpha k}^* U_{\beta k} U_{\alpha j} U_{\beta j}^*) \sin\left(\frac{\Delta m_{kj}^2 L}{2E}\right)$$

$$\bar{P}(\nu_\alpha \rightarrow \nu_\beta) = \delta_{\alpha\beta} - 4 \text{Re} \sum_{k,j=1} U_{\alpha k}^* U_{\beta k} U_{\alpha j} U_{\beta j}^* \sin^2\left(\frac{\Delta_{kj} L}{4E}\right) - 2 \text{Im} \sum_{k,j=1} U_{\alpha k}^* U_{\beta k} U_{\alpha j} U_{\beta j}^* \sin\left(\frac{\Delta_{kj} L}{2E}\right)$$

$$P_{\nu_\alpha \rightarrow \nu_\beta} - P_{\bar{\nu}_\alpha \rightarrow \bar{\nu}_\beta} = 4 \sum_{i>j} \mathcal{I}(U_{\beta i} U_{\alpha i}^* U_{\beta j}^* U_{\alpha j}) \sin\left(\frac{\Delta m_{ij}^2 L}{2E}\right)$$

- $\mathcal{I}(U_{\beta i} U_{\alpha i}^* U_{\beta j}^* U_{\alpha j}) = J \sum_{\gamma, l} \epsilon_{\alpha\beta\gamma} \epsilon_{jkl}$ This J term here is the Jarlskog

$$J = s_{13} c_{13}^2 s_{12} c_{12} s_{23} c_{23} \sin \delta$$

$$P_{\nu_{\alpha} \rightarrow \nu_{\beta}} - P_{\bar{\nu}_{\alpha} \rightarrow \bar{\nu}_{\beta}} = 8J \sin \left(\frac{\Delta m_{13}^2 L}{4E} \right) \sin \left(\frac{\Delta m_{23}^2 L}{4E} \right) \sin \left(\frac{\Delta m_{21}^2 L}{4E} \right)$$

The amount of CP-violation in a given oscillation channel $\nu_{\alpha} \rightarrow \nu_{\beta}$ with $\beta \neq \alpha$ can be found from the above equation.

Some Points and Important factors about this CP Violation:

1. CP violation is not possible for a two-neutrino system, because the most general mixing matrix has no complex parameters.
2. CP-violation is not possible in disappearance channels where we search for $\nu_{\alpha} \rightarrow \nu_{\alpha}$.
3. CP violation is only observable at baselines sufficiently long.

Wavepacket Approach in Neutrino Oscillations

Let us consider a neutrino with definite flavor α ($\alpha = e, \mu, \tau$), that propagates along x axis. We can write:

$$|v_\alpha(x, t)\rangle = \sum_j U_{\alpha j}^* \psi_j(x, t) |v_j\rangle, \quad (32)$$

$$\psi_j(p) = (2\pi\sigma_p^{P^2})^{-\frac{1}{4}} \exp\left\{-\frac{(p - p_j)^2}{4\sigma_p^{P^2}}\right\}$$

We consider this momentum wave function (Gaussian)

$$\sigma_x^P = \frac{1}{2\sigma_p^P}$$

$$\psi_j(x, t) = \frac{1}{\sqrt{2\pi}} \int dp \psi_j(p) e^{ipx - iE_j(p)t}$$

Solve the Fourier series

To solve this Fourier transform, we need to consider some approximations:

- Gaussian momentum wave function being strongly peaked around *i.e.* $\sigma_p^P \ll E_j^2(p_j)/m_j$
- Each mass eigenstates have different group velocities.

So finally with these approximations, we get the Fourier series and get :

$$\psi_j(x, t) = (2\pi\sigma_x^{P^2})^{-\frac{1}{4}} \exp\left[-iE_j t + ip_j x - \frac{(x - v_j t)^2}{4\sigma_x^{P^2}}\right]$$

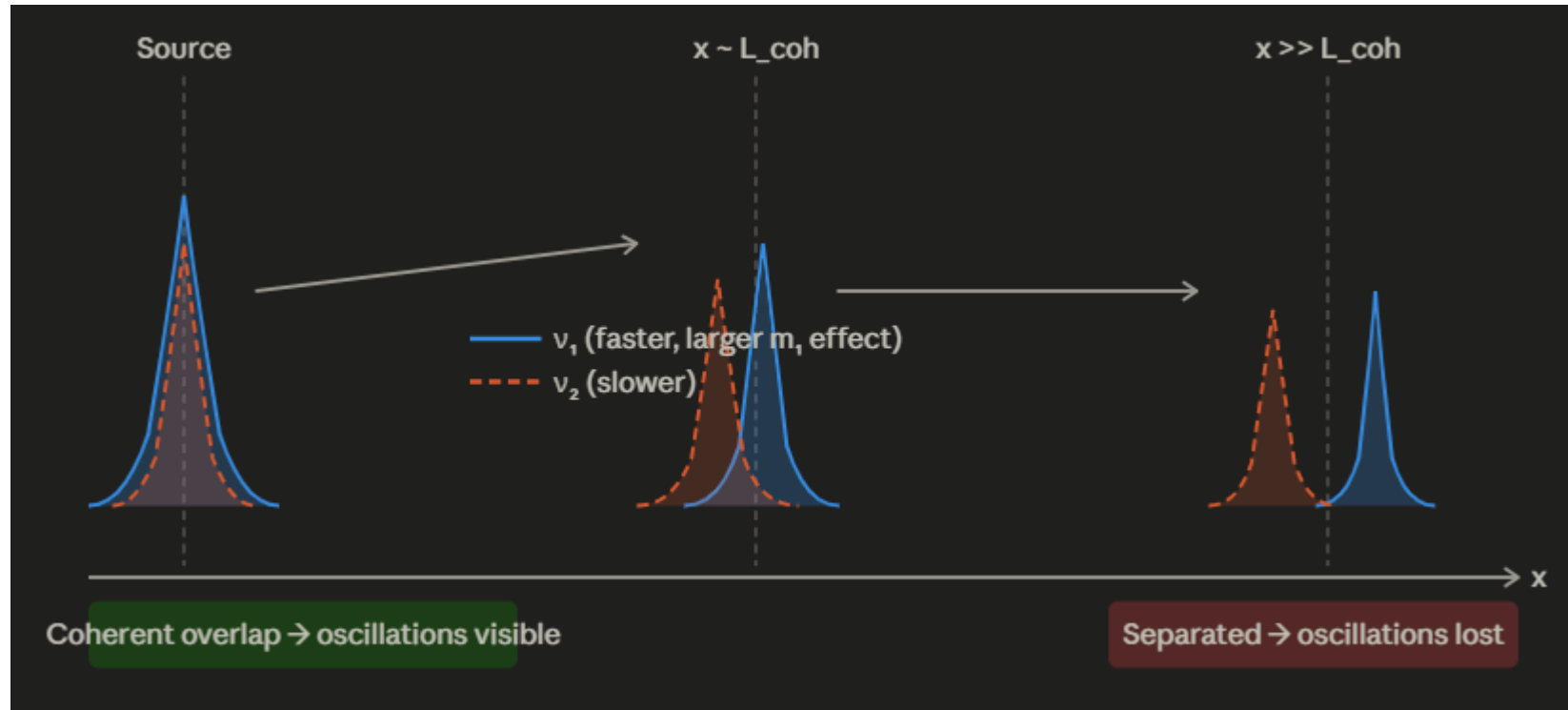
This is the Spatial Wavepacket

Now using that spatial width in the neutrino wave function, we get, the final probability of neutrinos, which is :

$$P_{\nu_\alpha \rightarrow \nu_\beta}(L) = \text{Tr}(\rho_\alpha(x) \mathcal{O}_\beta(x - L))$$

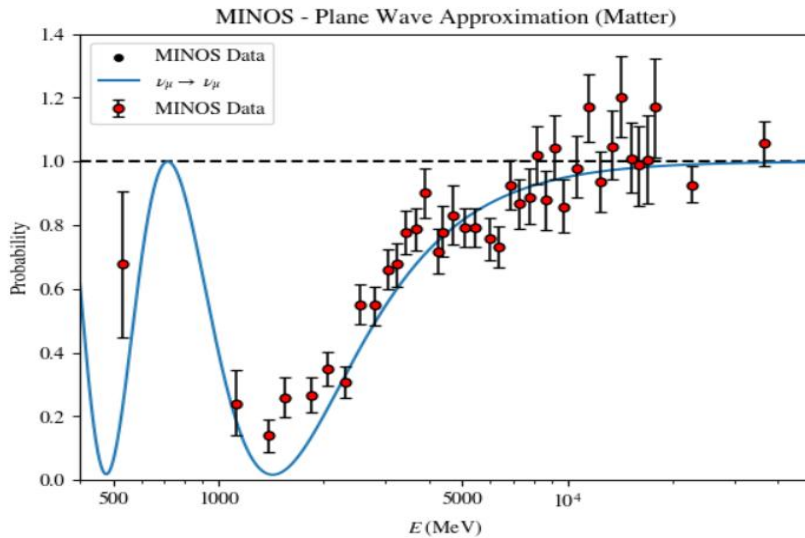
$$= \sum_{j,k} U_{\alpha j}^* U_{\alpha k} U_{\beta j}^* U_{\beta k} \exp \left[-2\pi i \frac{L}{L_{jk}^{osc}} - \left(\frac{L}{L_{jk}^{coh}} \right)^2 - 2\pi^2 (1 - \xi)^2 \left(\frac{\sigma_x}{L_{jk}^{osc}} \right)^2 \right]$$

Here, L_{osc} is the oscillation length and L_{coh} is the coherent length.



Physical meaning on how neutrino oscillation depends while moving from the source

Theoretical and Experimental Results:

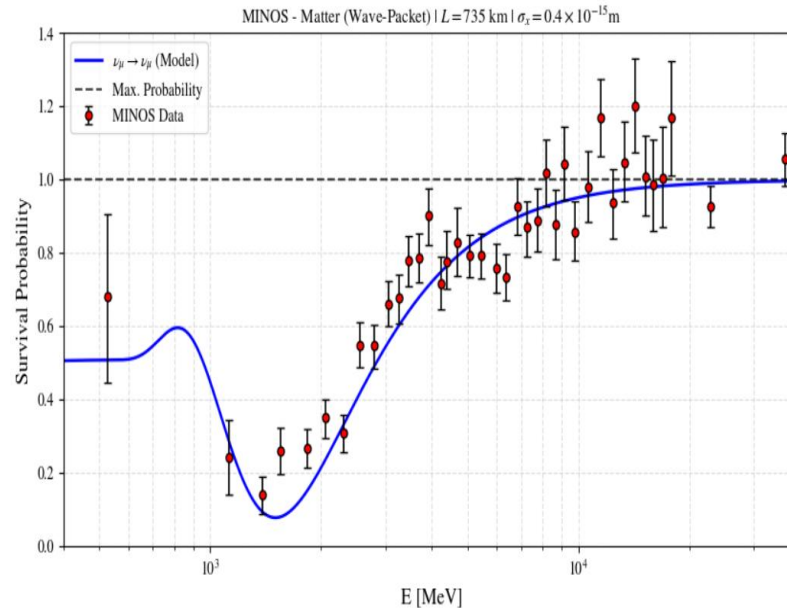


This are the theoretical results with the MINOS experiment's parameters,

We tried with both the approximations – Plane Wave and Wave-Packet approximations

$$\Delta m_{21}^2 \approx 7.39 \times 10^{-5} eV^2; \quad \Delta m_{31}^2 \approx 2.45 \times 10^{-3} eV^2;$$

$$L = 735 \text{ km} \quad \theta_{12} \approx 33.82; \quad \theta_{13} \approx 8.61; \quad \theta_{23} \approx 49.7.$$



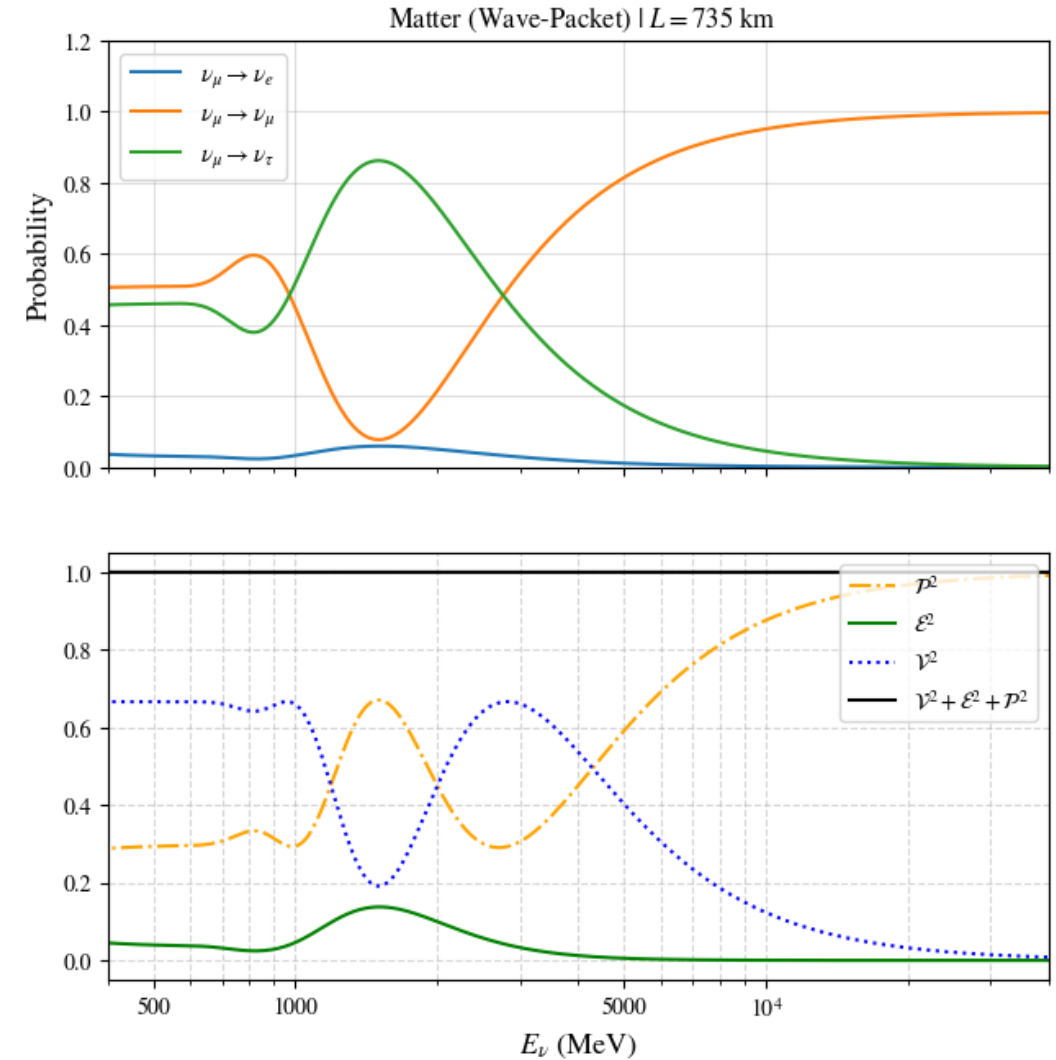
Fit with the experimental data

Future Ideas and Collaborations:

Our Next plan is to extend this work and study **Quantum Information Parameters** like **Predictability, Entanglement and Visibility** for **MINOS** and **DUNE** which are long baseline experiments.

This work is done under supervision of Prof. Massimo Blasone and Prof. Bipin Singh Koranga

$$\sigma_x \approx 0.4 \times 10^{-15} \text{ m (MINOS)}$$



SOME PICTURES OF THE GOOGLE MEETS REGARDING OUR DISCUSSION FOR THIS TOPIC.



AsiaEnergy Institute

Overview

The discovery of neutrino oscillations, implying for nonzero neutrino masses, and the existence of dark matter call for new physics beyond the Standard Model (SM) of elementary particles and cosmology. Fundamental questions regarding the origin of electroweak scale, the generation of neutrino masses and matter-antimatter asymmetry, the nature of dark matter and the dynamical evolution of the early Universe may find their answers in a new paradigm of particle physics. On the one hand, the next-generation neutrino oscillation experiments, with JUNO being in the successful operation and Hyper-Kamiokande and DUNE being under construction, bring us to the era of precision measurements at low energies. On the other hand, the Higgs factories (e.g., CEPC and FCC-ee), 100 TeV hadron colliders (e.g., SpHC and FCC-hh), and Muon colliders have been planned by the international high-energy physics community to explore new physics beyond the electroweak scale. Meanwhile, recent advancements in quantum information science, gravitational wave observations, and multi-messenger astronomy have further opened new avenues for probing new physics. Interdisciplinary interactions that integrate insights and tools from various fields may be crucial for uncovering more clues.

This workshop aims to bring together experimentalists and theorists from neutrino physics, collider physics and other closely related areas. Our goal is to facilitate the exchange of recent advancements and foster discussions on the pursuit of new physics.

Time: August 14-16, 2026

Venue:
Room 124, Multidisciplinary Building
Institute of High Energy Physics, Chinese Academy of Sciences
No. 198 Yuquan Road, Shijingshan District, Beijing, China

Contact:
✉ ICP90202@ihep.ac.cn
☎ 010-86003814

Timetable
Contribution List
Registration
Participant List
Registration Fee Payment
Accommodation
Travel Information
1. Venue and Location
2. Local Dining and Attractions
3. Payments in China
Previous NOPP

Massimo BLASONE

Baktiar

Conceptual Study of Neutrino Oscillations

Baktiar Wasir Farooq¹, Bipin Singh Koranga², and Ng. K Francis³

¹*Department of Physics, Kirori Mal College, University of Delhi*

²*Department of Physics, Kirori Mal College, University of Delhi*

³*School of Physical Sciences, Jawaharlal Nehru University*

Abstract

In this paper we would try to cover the basics of Neutrino oscillations starting from 2 flavors to 3 flavors. Initially we will cover the history behind the Ideas of Neutrinos, then go forward to talk about the properties of neutrinos and it's general formalisms. This would give an idea of what exactly neutrinos are and how do they change flavor over time.

Last Semester's work under supervision of Prof. Francis and Prof. Bipin was been published in **The Physics Insights**.
(A new journal for research works published by RAMJAS COLLEGE)

RECENT WORKS DONE BY ME:

Quantum Fisher Information as a Probe of Sterile Neutrino New Physics:Geometric Advantage of KM3NeT over IceCube #1

Baktiar Wasir Farooq (Delhi, Kirori Mal Coll.), Bipin Singh Koranga (Delhi, Kirori Mal Coll.), Aritro Chatterjee (Delhi, Kirori Mal Coll.) (Apr 1, 2026)

e-Print: [2604.01256](#) [hep-ph]

 pdf  cite  claim

 reference search  0 citations

NeutrinoOsc3Flavor: CP Phase Dependence in Three-Flavor Neutrino Oscillations: A Numerical Study in Vacuum and Matter #2

Baktiar Wasir Farooq (Delhi, Kirori Mal Coll.), Bipin Singh Koranga (Delhi, Kirori Mal Coll.), Ansh Prasad (Delhi, Kirori Mal Coll.), Imran Khan (Ramjas Coll.) (Jan 31, 2026)

e-Print: [2602.02578](#) [hep-ph]

 pdf  cite  claim

 reference search  0 citations

Modified entanglement patterns in two-flavor neutrinos from quantum-gravity interactions #3

Bipin Singh Koranga (Delhi, Kirori Mal Coll.), Baktiar Wasir Farooq (Delhi, Kirori Mal Coll.), Pranav Kumar (Delhi, Kirori Mal Coll.) (Nov 29, 2025)

Published in: *Nucl.Phys.B* 1025 (2026) 117373 • e-Print: [2512.02068](#) [hep-ph]

 pdf  DOI  cite  claim

 reference search  3 citations

Two Flavour Neutrino Oscillation in Matter and Quantum Entanglement #4

Bipin Singh Koranga (Delhi, Kirori Mal Coll.), Baktiar Wasir Farooq (Delhi, Kirori Mal Coll.) (Oct 11, 2024)

e-Print: [2410.09137](#) [hep-ph]

 pdf  DOI  cite  claim

 reference search  1 citation

**Thank you so much for your
valuable time**